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Subject : Machine Learning Lab

Name: _____

Roll No. : _____

Class: T.Y. Comp Engg.

Batch : T

Division: _____

Expt. No. : 04

Performance Date : / /

Remark

Signature

Title : Write a program to demonstrate the working of the decision tree based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample and Visualize model with Python.

Rubrics:

Date of Submission:		
Marks Split up to	Maximum Marks	Marks Obtained
Performance/conduction		
Report Writing		
Attendance		
Viva/Oral		
Total Marks		

Aim :

The main aim of this lab is to demonstrate the how to Creating and Visualizing Decision Trees with Python

Secondary aim of this lab is to undersand

- What Decision Tree is
- What Decision Tree is used for
- How to Fit a Decision Tree Model using Scikit-Learn
- How to Visualize Decision Trees using Matplotlib

Software Required :

Jupyter Notebook , Python 3.6, Or Google Colab

Dataset Used:

bill_authentication.csv

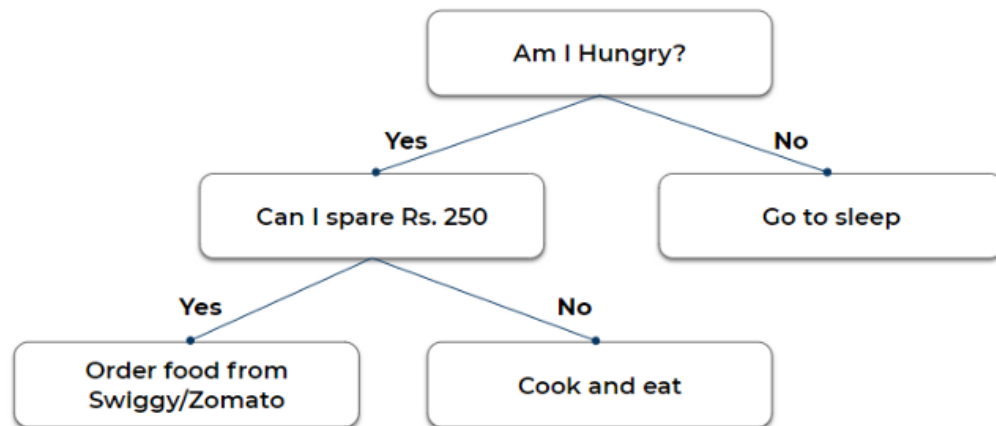
Theory :

Introduction to Decision Tree

Formally a decision tree is a graphical representation of all possible solutions to a decision. These days, tree-based algorithms are the most commonly used algorithms in case of supervised learning scenarios. They are easier to interpret and visualize with great adaptability. We can use tree-based algorithms for both regression and classification problems, However, most of the time they are used for classification problem.

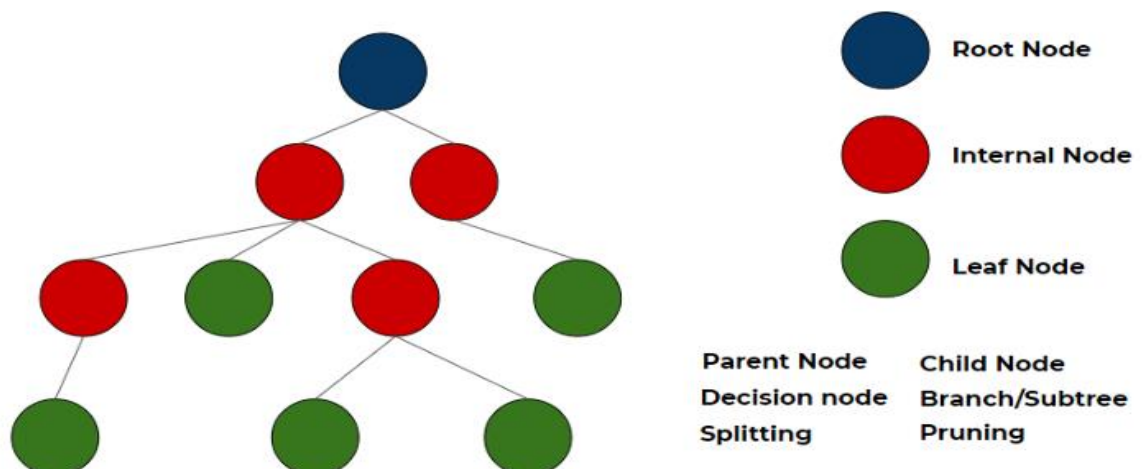
Let's understand a decision tree from an example: Yesterday evening, I skipped dinner at my usual time because I was busy taking care of some stuff. Later in the night, I felt butterflies in my stomach. I thought only if I wasn't hungry, I could have gone to sleep as it is but as that was not the case, I decided to eat something. I had two options, to order something from outside or cook myself.

I figured if I order, I will have to spare at least INR 250 on it. I finally decided to order it anyway as it was pretty late and I was in no mood of cooking. This complete incident can be graphically represented as shown in the following figure.



This representation is nothing but a decision tree.

Terminologies associated with decision tree



- **Parent node:** In any two connected nodes, the one which is higher hierarchically, is a parent node.
- **Child node:** In any two connected nodes, the one which is lower hierarchically, is a child node.
- **Root node:** Starting node from which the tree starts, It has only child nodes. The root node does not have a parent node. (dark blue node in the above image)
- **Leaf Node/leaf:** Nodes at the end of the tree, which do not have any children are leaf nodes or called simply leaf. (green nodes in the above image)
- **Internal nodes/nodes:** All the in-between the root node and the leaf nodes are internal nodes or simply called nodes. internal nodes have both a parent and at least one child. (red nodes in the above image)
- **Splitting:** Dividing a node in two or more sub-nodes or adding two or more children to a node.
- **Decision node:** when a parent splits into two or more children nodes then that node is called a decision node.
- **Pruning:** When we remove the sub-node of a decision node, it is called pruning. You can understand it as the opposite process of splitting.
- **Branch/Sub-tree:** a subsection of the entire tree is called a branch or sub-tree.

Types of Decision Tree

1. Regression Tree

A regression tree is used when the dependent variable is **continuous**. The value obtained by leaf nodes in the training data is the **mean** response of observation falling in that region. Thus, if an unseen data observation falls in that region, its prediction is made with the mean value. This means that even if the dependent variable in training data was continuous, it will only take discrete values in the test set. A regression tree follows a **top-down greedy approach**.

2. Classification Tree

A classification tree is used when the dependent variable is **categorical**. The value obtained by leaf nodes in the training data is the **mode** response of observation falling in that region. It follows a **top-down greedy approach**.

Building a decision Tree from data

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
Id					
1	5.1	3.5	1.4	0.2	Iris-setosa
2	4.9	3.0	1.4	0.2	Iris-setosa
3	4.7	3.2	1.3	0.2	Iris-setosa
4	4.6	3.1	1.5	0.2	Iris-setosa
5	5.0	3.6	1.4	0.2	Iris-setosa



How to create a tree from tabular data? which feature should be selected as the root node? on what basis should a node be split? To all these questions answer is in this section

The decision of making strategic splits heavily affects a tree's accuracy. The purity of the node should increase with respect to the target variable after each split. The decision tree splits the nodes on all available variables and then selects the split which results in most homogeneous sub-nodes.

The following are the most commonly used algorithms for splitting

1. Gini impurity

Gini says, if we select two items from a population at random then they must be of the same class and the probability for this is 1 if the population is pure.

1. It works with the categorical target variable "Success" or "Failure".
2. It performs only Binary splits
3. Higher the value of Gini higher the homogeneity.
4. CART (Classification and Regression Tree) uses the Gini method to create binary splits.

Steps to Calculate Gini impurity for a split

1. Calculate Gini impurity for sub-nodes, using formula subtracting the sum of the square of probability for success and failure from one.
 $1-(p^2+q^2)$
where $p = P(\text{Success})$ & $q = P(\text{Failure})$
2. Calculate Gini for split using the weighted Gini score of each node of that split
3. Select the feature with the least Gini impurity for the split.

2. Chi-Square

It is an algorithm to find out the statistical significance between the differences between sub-nodes and parent node. We measure it by the sum of squares of standardized differences between observed and expected frequencies of the target variable.

1. It works with the categorical target variable "Success" or "Failure".
2. It can perform two or more splits.
3. Higher the value of Chi-Square higher the statistical significance of differences between sub-node and Parent node.
4. Chi-Square of each node is calculated using the formula,
5. $\text{Chi-square} = ((\text{Actual} - \text{Expected})^2 / \text{Expected})^{1/2}$
6. It generates a tree called CHAID (Chi-square Automatic Interaction Detector)

Steps to Calculate Chi-square for a split:

1. Calculate Chi-square for an individual node by calculating the deviation for Success and Failure both

2. Calculated Chi-square of Split using Sum of all Chi-square of success and Failure of each node of the split
3. Select the split where Chi-Square is maximum.

3. Information Gain

A less impure node requires less information to describe it and, a more impure node requires more information. Information theory is a measure to define this degree of disorganization in a system known as Entropy. If the sample is completely homogeneous, then the entropy is zero and if the sample is equally divided (50% — 50%), it has an entropy of one. Entropy is calculated as follows.

$$Entropy = -p.\log_2^p - q.\log_2^q$$

Steps to calculate entropy for a split:

1. Calculate the entropy of the parent node
2. Calculate entropy of each individual node of split and calculate the weighted average of all sub-nodes available in the split. The lesser the entropy, the better it is.
3. calculate information gain as follows and chose the node with the highest information gain for splitting

$$Information\ Gain = 1 - Entropy$$

4. Reduction in Variance

Till now, we have discussed the algorithms for the categorical target variable. Reduction in variance is an algorithm used for continuous target variables (regression problems).

1. Used for continuous variables
2. This algorithm uses the standard formula of variance to choose the best split.
3. The split with lower variance is selected as the criteria to split the population

Steps to calculate Variance:

1. Calculate variance for each node.

$$\sigma^2 = \frac{\sum_{i=1}^N (x_i - \mu)^2}{N}$$

1. Calculate variance for each split as a weighted average of each node variance.
2. The node with lower variance is selected as the criteria to split.

In summary:

Gini says, if we select two items from a population at random then they must be of same class and probability for this is 1 if population is pure.

Reduction in variance is an algorithm used for continuous target variables (regression problems). This algorithm uses the standard formula of variance to choose the best split.



to find out the statistical significance between the differences between sub-nodes and parent node.

Less impure node requires less information to describe it and, more impure node requires more information.

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Procedure: Implementing Decision Tree in Python

Step-1: Importing the packages

Our primary packages involved in building our model are pandas, scikit-learn, and NumPy.

Step-2: Importing data and EDA

In this step, we will be utilizing the 'Pandas' package available in python to import and do some EDA on it. The dataset we will be using to build our decision tree model is a [bill_authentication](#)

Now we have a clear idea of our dataset. After importing the data, let's get some basic information on the data using the 'info' function. The information provided by this function includes the number of entries, index number, column names, non-null values count, attribute type, etc.

Step-3: Data Processing

Step-4: Splitting the data

After processing our data to be of the right structure, we are now set to define the 'X' variable or the independent variable and the 'Y' variable or the dependent variable.

Step-5: Building the model & Predictions

Building a decision tree can be feasibly done with the help of the 'DecisionTreeClassifier' algorithm provided by the scikit-learn package. After that, we can make predictions of our data using our trained model. Finally, the precision of our predicted results can be calculated using the 'accuracy_score' evaluation metric.

Step-6: Visualizing the model

Now that we have our decision tree model and let's visualize it by utilizing the 'plot_tree' function provided by the scikit-learn package in python. Follow the code to produce a beautiful tree diagram out of your decision tree model in python.

Task to Perform:

1. Implement Decision tree and find the accuracy of your model
2. Visualize your build model in python

Conclusion

In this way we successfully implement and visualize Decision Tree in

Test your Knowledge

1. For the model we created in lab session write the following results

Accuracy	
Precision	
Recall	
F1 Score	

2. Create the model with following settings

- a. **Criterion = *log_loss***
- b. **Splitter = *random***
- c. **max_features = *sqrt***
- d. **random_state = any number of your choice**

write a code in your .ipynb file and find following result

Accuracy	
Precision	
Recall	
F1 Score	